

Tuesday Warm-up: Biot-Savart and Ampère's Laws

Prof. Jordan C. Hanson

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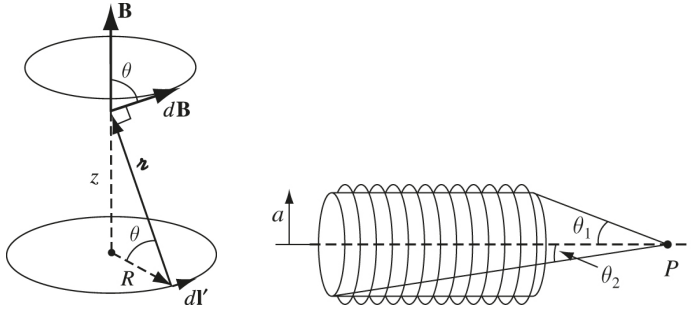


Figure 1: (Left) The $\mathbf{B}$ -field above a circular loop of current. (Right) A solenoid.

2. Consider Fig. 1 (right). Find the magnetic field at point P on the axis of a tightly wound solenoid consisting of n turns per unit length wrapped around a cylindrical tube of radius a and carrying current I . *Hint: express your answer in terms of θ_1 and θ_2 , and let $I \rightarrow nIdz$ in the Biot-Savart law.*

1 Memory Bank

- **The Biot-Savart Law**, for steady currents:

$$\mathbf{B}(\mathbf{r}) = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}' \times \hat{\mathbf{r}}}{r^2} \quad (1)$$

- **Ampère's Law**, differential form:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad (2)$$

- **Ampère's Law**, integral form:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} \quad (3)$$

2 Biot-Savart Law

1. Consider Fig. 1 (left). Find the magnetic field a distance z above the center of a circular loop of radius R , which carries a steady current I .

3 Ampère's Law

1. Recall the result for the $\mathbf{B}$ -field a distance s for a long, straight wire:

$$\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi} \quad (4)$$

- (a) Calculate the line integral of $\mathbf{B}$ along a circular path surrounding the wire:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \dots \quad (5)$$

- (b) Suppose this applies to *any* geometry, and not just a straight wire. How can you use this result to derive the field of the solenoid?